fungal infection
anti fungal drug

Dr eman gomaa

Consultant dermatologist

Taxonomy

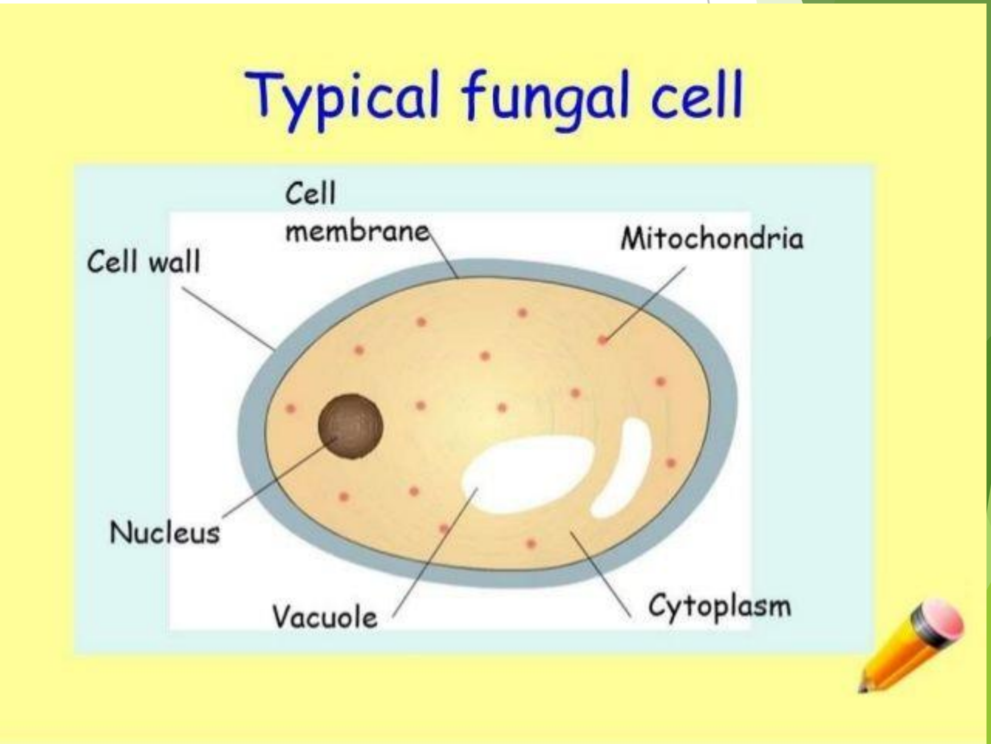
Separate kingdom

Eukaryotic

Structure:-

Nutrition :- Not autotrophs

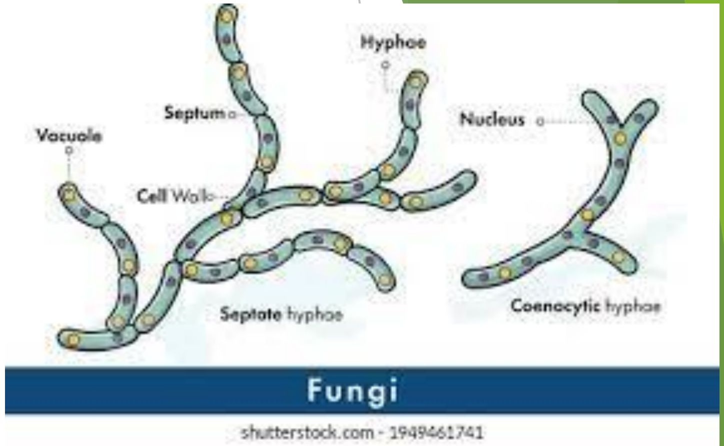
1. Saprophytes.
2. Parasites.
3. Symbiotic.



Fungal morphology

mould	yeast	dimorphic

Mould



- Multicelluar long nucleated filaments.

- Hyphae
  - Separated
  - Aseparated :- genocytic all nuclei lie together in big undevied mass of cytoplasm.

Growth:- hyphae grow at apical tip.

Hyphae branch regularly.

Mycellium :- aggregation of hyphae

Thallus :- whole moys of fungi

Culture: cottony , velvety colonies

Reproduction

- Sporulation, budding and binary fission
  - Sporulation :- Sexual & Asexual

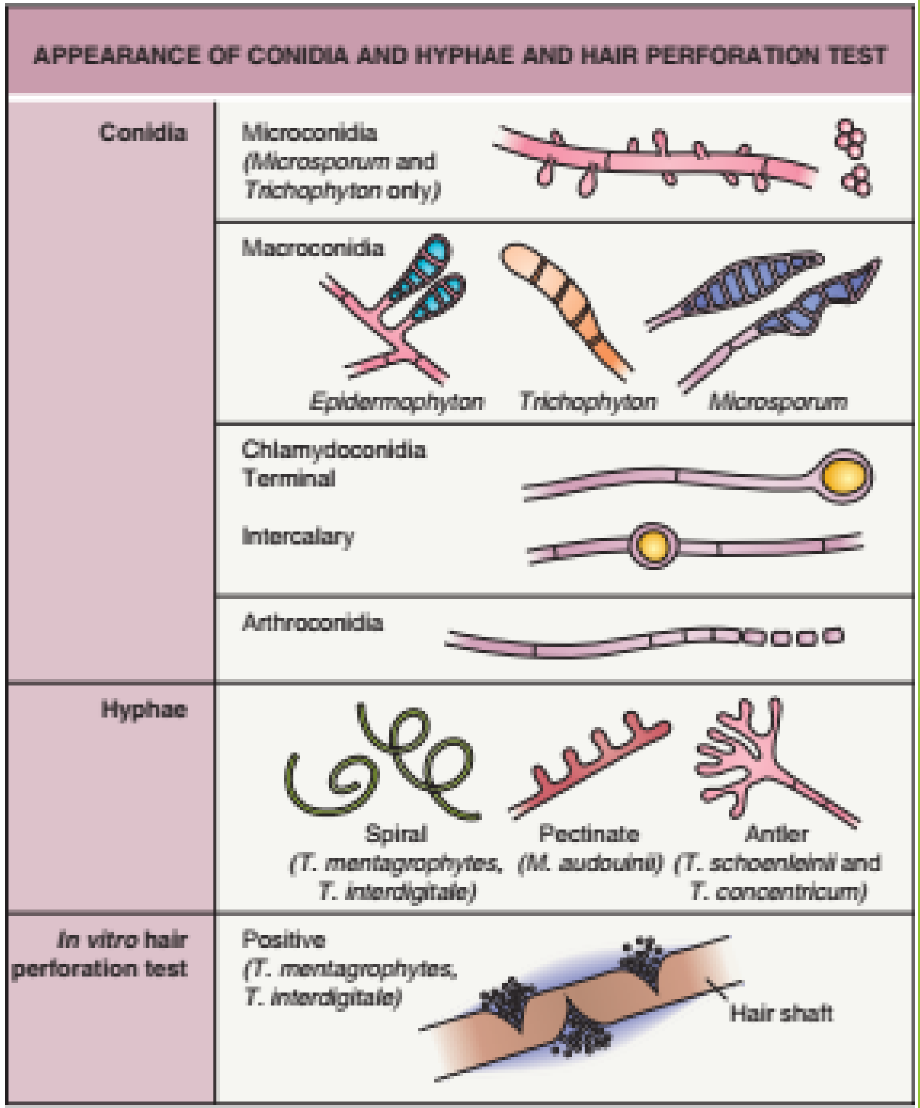


Fig. 77.19 Microscopic appearance of various forms of conidia and hyphae and the *in vitro* hair perforation test.

yeast

- Rounded or oval unicellular cell
- Reproduction:-

1. Budding
2. Fission.
3. Filaments :- true and pseudo hyphae → constriction, shortening and branching.

Culture:- soft pasty or mucoid colonies.

Dimorphic

Depend on environmental condition

Temperature

Nutrition

Moulds → yeast.

- Malassezia

Yeast from saprophytes.

Hyphae → parasitic.

- Histoplasma capsulatum

25 cm → moulds

37 cm → yeast

Diagnosis of fungal infection

- Basic procedure :-

Wood's lamb.

Microscopic → koh 10-20 %

chlorazol black & calcoflur.

- Advanced:-

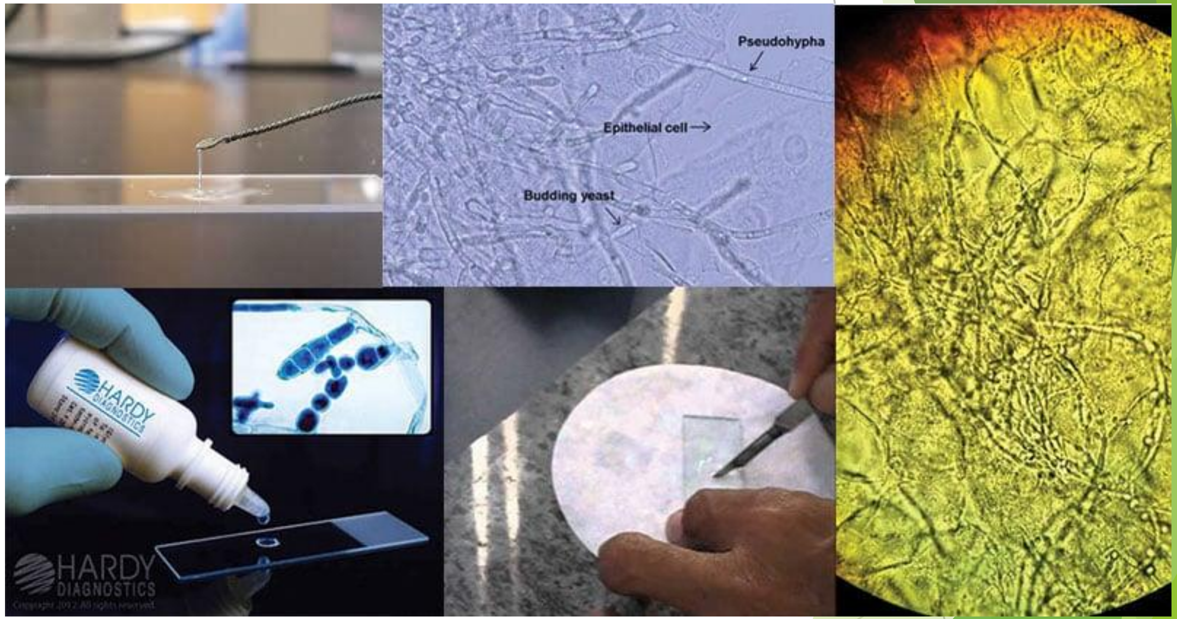
Biopsy

→ PAS-GMS- Fontana magon

→Mayer's mucormine

→ Fire stain

PCR, Biochemical & Dermoscopy.


Pseudohypha
Epithelial cell
Budding yeast
HARDY DIAGNOSTICS
HARDY
DIAGNOSTICS

PROPER SPECIMEN COLLECTION

Skin specimens
- Cleanse skin with alcohol or soap/water and allow to dry
- Scrape scale from the advancing border of lesion with no.15 blade or glass slide

Hair specimens
- Epilate several broken hairs with tweezers; if Wood's lamp examination is performed, remove any hairs that fluoresce
- Scrape scale from affected scalp with a blade
- For scalp culture, an option is to swab affected scalp with a cotton tip applicator or culturette*

Nail specimens
- Cleanse affected nail(s) with alcohol or soap/water and allow to dry
- Clip nail(s) to the most proximal point possible (without causing discomfort)
- Collect any subungual debris by scraping the area under trimmed nail with 1–2 mm serrated curette or no.15 blade

- Perform KOH examination of specimens or PAS if nail specimen
- Place scale, hairs and/or nails on culture media (Sabouraud dextrose agar with chloramphenicol and cycloheximide +/– plain Sabouraud dextrose agar, depending on presumed organism)
- If sending to fungal reference library place samples in a sterile container

*If culturette is used for transport, do not break the ampule; swab fungal culture media with applicator; KOH/calcofluor are not possible with this method

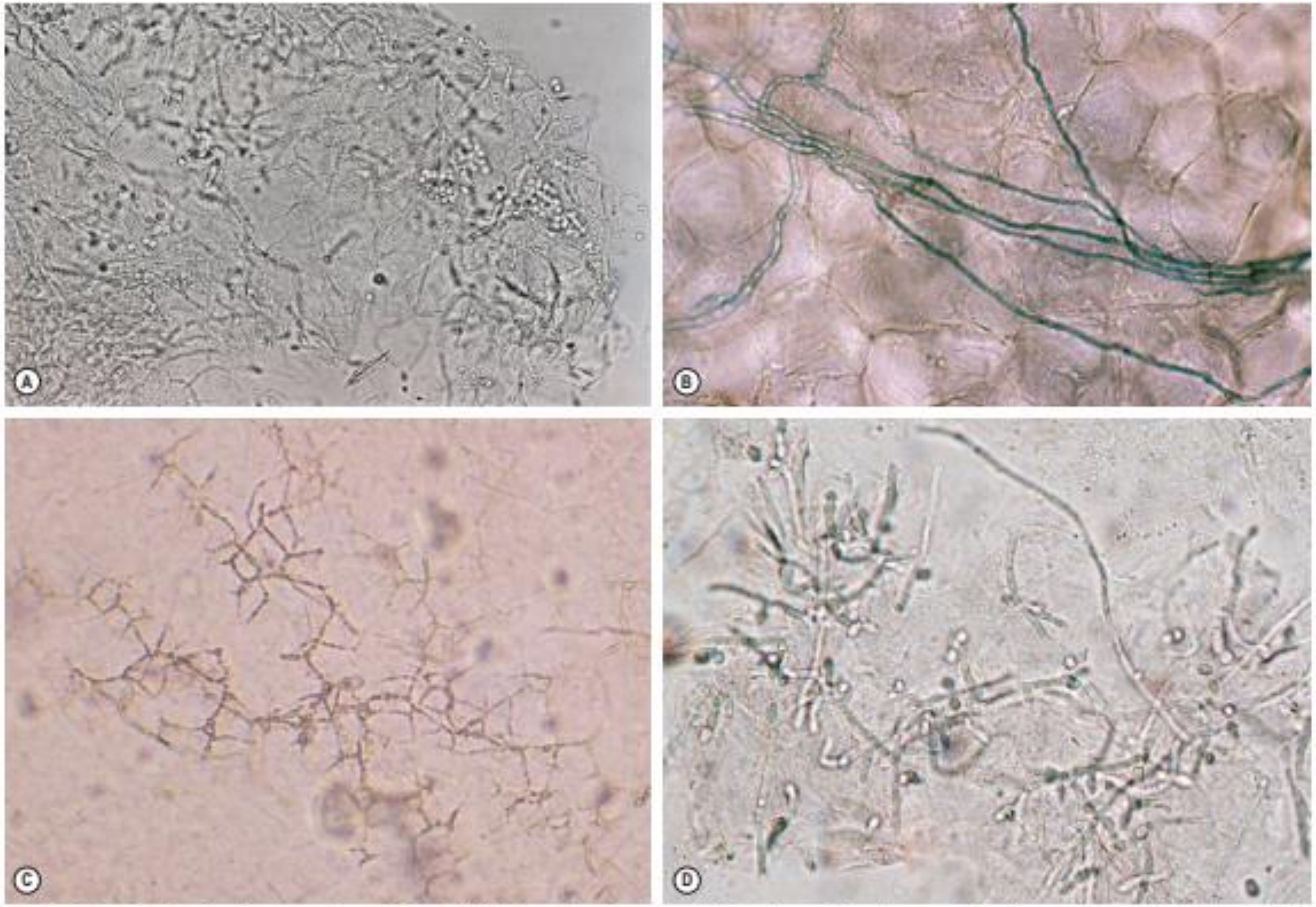


Fig. 77.5 Potassium hydroxide preparations. A Superficial skin scrapings from tinea (pityriasis) versicolor demonstrating clusters of yeast and short mycelial forms. **B** A dermatophyte, in this case *Trichophyton tonsurans*, demonstrating branching hyphae (chlorazol black stain). Note that the hyphae cross multiple squamous cells. **C** "Mosaic" pattern of cell walls that should not be misinterpreted as hyphae. **D** Yeast and pseudohyphae of candidiasis. *A, Courtesy, Ronald P Rapini, MD; C, Courtesy, Louis A Fragola, Jr, MD; D, Courtesy, Frank Samarin, MD.*

Antifungal drugs

acetat

↓

squalene

↓

Allylamines
terbinafine → Squalene epoxidase

↓

lanosterol

↓

Azoles

- Trizoles
Fluconazol
Itraconazol

- Imidazole
ketoconazole

→ 14 alpha demethylase

↓

14 alpha demethylase
lanesterol

↓

Ergosterol →

Nystatin &
amphotericin B

Polyenes
→ bind to ergosterol
→ leakage cellular content → cell death

Griseofulvin
Act on cell devision
(mitotic spindl)

- Griseoflavin → act on cell division (mitotic spindle).

- Choice of systemic anti-fungal

Terbinafen →dermatophyte.

Griseoflavin → dermatophyte (tinea capitis).

Itraconazol → broad spectrum.

Fluconazol → candida.

4) Flucytosine →5 flurouracil→ DNA syntgesis. ↓

5) Echinocardins → alter cell wall permeability.

Antifungal

Terbinafine

- Chemistry:- synthetic allylamine.
- Kinetic:- increased absorption with fat & acidic food
- Distribution:- sebum, skin & nail.

  - Action:-

✓ Fungicidal:- ↑ squaline → toxic.

✓ Fungistatic :- ergosterol.

- ***Mainly against dermatophyte***

Griseoflavin

- Penicillum griseoflavin (antifungal & antibiotic).
- Fatty meal.
- Keratin & ecrine sweat.
- Fungistatic:↓ mitotic spindles →

Fungal mitosis↓

Dermatophyte only

Terbinafine

- Dose:- 125 – 250 mg
- 10-20kg → 62
- 20-40 kg → 125
- >40 kg→ 250
- 2-6 weeks (tinea)

Griseoflavin

- 125 mg tab & syrup
- 12.5 mg/kg/day ultramicrosize
- FDA approved in tinea capitis (6-12 weeks

Anti-inflammatory (anti-tumor)

LP (Oral

Terbinafine

- Side effects:-
- Gastric upset, metallic taste
- SCLE, PS, SLE & urticarial
- Category B in pregnancy.
- Not used in lactation.
- Children < 4 ys.

Griseoflavin

- Headache, VD
- GIT
- Photosensitivity → Subacute LE
- Category C in pregnancy.

- # Terbinafine

- Interaction:-

Rifampicin → ↓ terbinafine.

Cimetidine→ increase terbinafine.

Not with OCP.

Griseoflavin

- Alcohol.
- OCP → failure.
- Barbiturate :- ↓ griseoflavin

- Contraindications

L.E.

Porphyria

)

Itraconazole

- Triazole
- increased absorption with fat & acidic food.
- Distributed :- sebum, sweat by passive diffusion.
- Fungistatic only (broad spectrum).
- Dose :- 100 mg cab BID (Pulse therapy)
  - S/E:-
- GIT upset. Pregnancy category C. Children <12 ys.
- Rare:- S.E:- hepatotoxicity & cardiac failure.
- AGEB, SjS, PS-urticaria

- Contraindicated with CCB → heart failure.
- Antacids:- decreased absorption.
- Rifampicin :- ↓

Fluconazole

- Triazole
- No fat, No acidic food & water soluble.
- Distribution:- direct diffusion (fungistatic).
- Mainly candida.
- Systemic , oropharyngeal & esophageal.
- 150 mg cap
- 150 mg /week.
- Onychomycosis → 300 mg / wk. → 4-8 months.
- Same side effects like itraconazole.

- Interactions:-
- ✓ Rifampicin → $\downarrow$ its level.
- ✓ Hydroxychlorothiazide → $\uparrow$ its level.
- ✓ Fluconazole → $\uparrow$ phenytoin, Cyclosporin.

Excreted unchanged in urine.

Used in neonates.

Pregnancy category C.

ORGANIZATION OF CUTANEOUS MYCOSES	
Superficial	Involve stratum corneum, hair, or nails
Subcutaneous	Involve dermis or subcutaneous tissue Often due to implantation
Systemic ("deep")	Involve dermis or subcutaneous tissue
"True" pathogens	Skin involvement usually reflects hematogenous spread or extension from underlying structures
Opportunistic	Primary or secondary skin lesions in immunocompromised hosts

SUPERFICIAL MYCOSES OF THE SKIN		
	Cutaneous disorder	**Pathogen(s)**
Minimal, if any, inflammation	Tinea (pityriasis) versicolor Tinea nigra Black piedra White piedra	*Malassezia furfur*, *M. globosa* *Hortaea werneckii* *Piedraia hortae* *Trichosporon* spp.*, e.g. *T. ovoides* (especially of scalp hair), *T. inkin* (especially of pubic hair), *T. cutaneum*, *T. loubieri*
Inflammatory response common	Tinea capitis, barbae, faciei, corporis, cruris, manuum, pedis Cutaneous candidiasis	*Trichophyton*, *Microsporum*, *Epidermophyton* spp. *Candida albicans*, other *Candida* spp.

Superficial cutaneous fungal infection

Moulds

Yeasts

Yeast like Malassezia

Dermatophyte

Non Dermatophyte

Others

- **Human**
  Furur
  Obtusa
  Glubosa
  Sleoffiae
  Sympodia
  Restrica
  Dermatis
  Japonica
  yamalenosis

- **Zophilic**
  Nona
  Caprae
  Equina
  Cunical
  Pachydermatis
  (Non lipid)

Dermatophyte

Microsporum

Trichophyton

Epidermophyton

DERMATOPHYTES ISOLATED WORLDWIDE		
		Thallus (macroscopic) appearance* and/or microscopic findings
Most common		
Trichophyton	*mentagrophytes* (previously *mentagrophytes* var. *mentagrophytes*)	Granular front, buff reverse; pencil-shaped macroconidia, clusters of round microconidia, spiral hyphae
	interdigitale (previously *mentagrophytes* var. *interdigitale*)	Downy front, buff reverse; see above
	rubrum	White wooly front, venous blood reverse; pencil-shaped macroconidia, teardrop-shaped microconidia
	tonsurans	Granular front, mahogany reverse; pencil-shaped macroconidia, microconidia of varying sizes
	verrucosum	Convoluted, cream to gray, compact; chains of chlamydospores at 37°C
	violaceum	Creamy, waxy, becomes violet
Microsporum	*canis*	White wooly front, orange reverse; multi-celled, spindle-shaped macroconidia with thick walls and rough surface
	ferrugineum	Folded red-orange (rust-colored) front
	gypseum	Cinnamon-tan granular front; multi-celled, cucumber-shaped macroconidia with thin walls
Epidermophyton	*floccosum*	Khaki green, suede to granular; beaver tail-shaped macroconidia; no microconidia
Less common		
Trichophyton	*ajelloi*	Powdery surface, resembles *Microsporum* spp.
	concentricum	Glabrous colonies; antler-like hyphae
	equinum	Club-shaped macroconidia
	gourvilii	Waxy, pink to red front
	megninii	Pink, felt-like front with red reverse
	schoenleinii	Glabrous; antler- and nailhead-like hyphae; rat-tail-like macroconidia (media often fissured)
	simii	Club-shaped macroconidia in clusters
	soudanense	Yellow to apricot front with fringed border
	terrestre	Cream to yellow granular surface
	yaoundei	Glabrous, chocolate-brown front
Microsporum	*amazonicum*	Multi-celled, spindle-shaped, macroconidia with large inclusions
	audouinii	Flat, tan front with salmon reverse; pectinate (comb-like) hyphae
	cookei	Oval, thick-walled macroconidia
	equinum	One- to four-celled macroconidia resembling *M. canis*
	fulvum	Bullet-shaped macroconidia with spiral hyphae
	gallinae	Diffusable pink-red pigment
	nanum	Two-celled macroconidia
	persicolor	Pink to red front and reverse, resembles *T. mentagrophytes*
	praecox	Powdery front with yellow-orange reverse
	racemosum	Cream-colored powdery front
	vanbreuseghemii	Largest macroconidia

Non dermatophyte

- Scytalidium dimidiatum.
- Aspergillus & fusarium.

- Others

Tinea nigra.

Piedra

Factors affecting infection

- Fungal
- Host

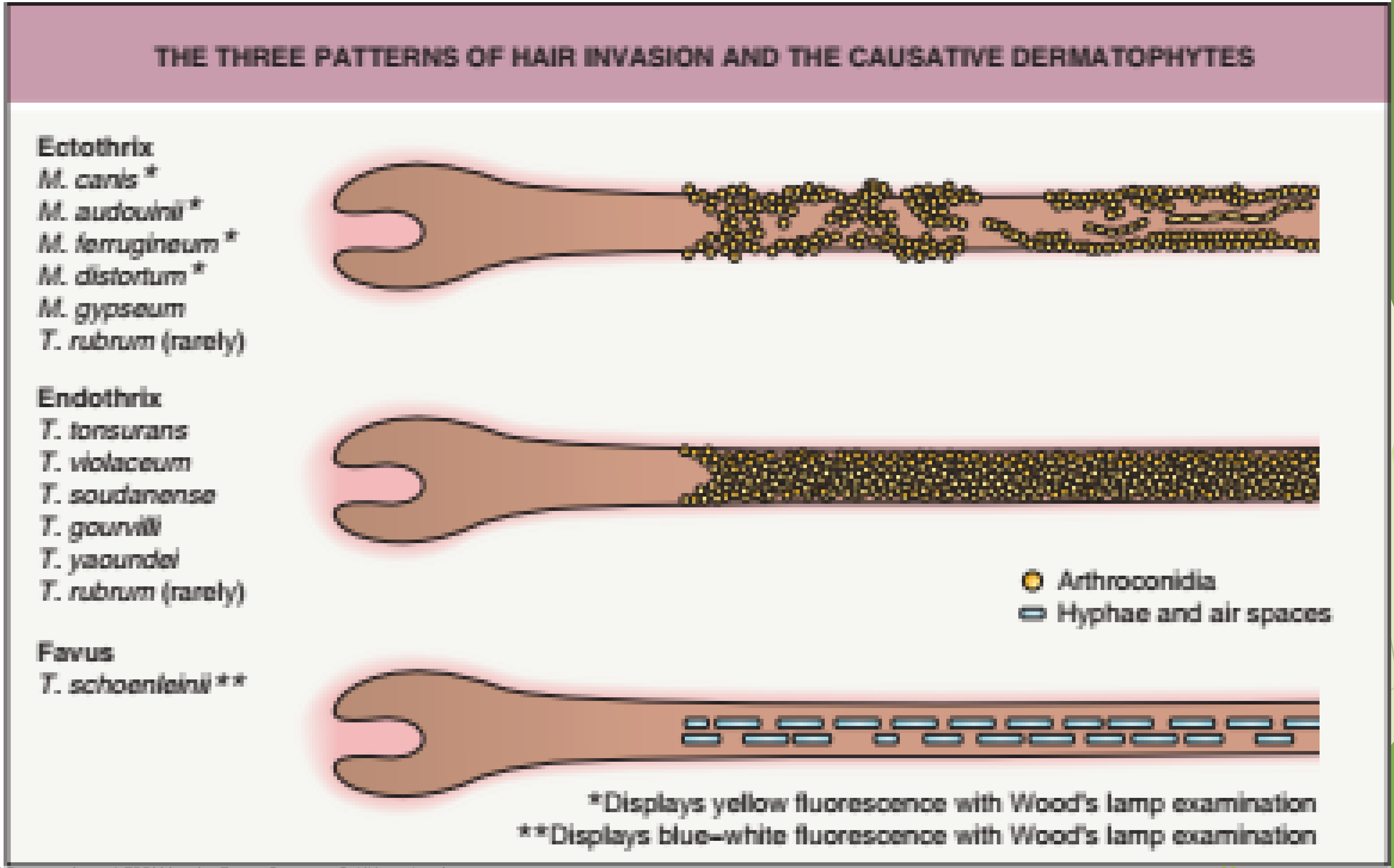
Tinea Capitis:-

Dermatophytic infection of scalp skin, hair follicle
Source of infection:- anthropophilic, geophyilic & zoophilic

- Causative:- microsporum trichophyton (Never epidermophyton).
- **Ectothrix:-** M, canis, M.aedoiini, T. verrucosum & T.mentographyte.

  invade hair follicle from adjacent S.C. , extrapillary hyphae ,

- **Endothrix:**- arthocondia inside hair shaft → T.tonsurans , T.violaceum TVS, soudaneses & rubrum →
- **favus :** →T.Schoenleinii → hyphea , air spaces
- few hyphae not break up into orthocondia, run intact → make a tunnel.
- Epidemiology:- more common in children.


THE THREE PATTERNS OF HAIR INVASION AND THE CAUSATIVE DERMATOPHYTES
Ectothrix
M. canis *
M. audouinii *
M. ferrugineum *
M. distortum *
M. gypseum
T. rubrum (rarely)
Endothrix
T. tonsurans
T. violaceum
T. soudanense
T. gourvilii
T. yaoundei
T. rubrum (rarely)
Favus
T. schoenleinii **
Arthroconidia
Hyphae and air spaces
*Displays yellow fluorescence with Wood's lamp examination
**Displays blue–white fluorescence with Wood's lamp examination

C.pic:-

Alopecia, scaling, inflammation & lymphadenopathy.

- Clinical variants

- ✓ **Scaly** →hair brake above skin surface → grey patch, fine scales ,ectothrix
- ✓ **Black dots**→ hair brake at skin surface →endothrex.
- ✓ **favus** → scatula, cicatrecial A → thick yellow cup-shaped crusts.
- ✓ **Kerion** →exaggerated resistance.

  Zoophilic painful-inflammatory boggy swelling with follicular discharge → sinus.


A


B

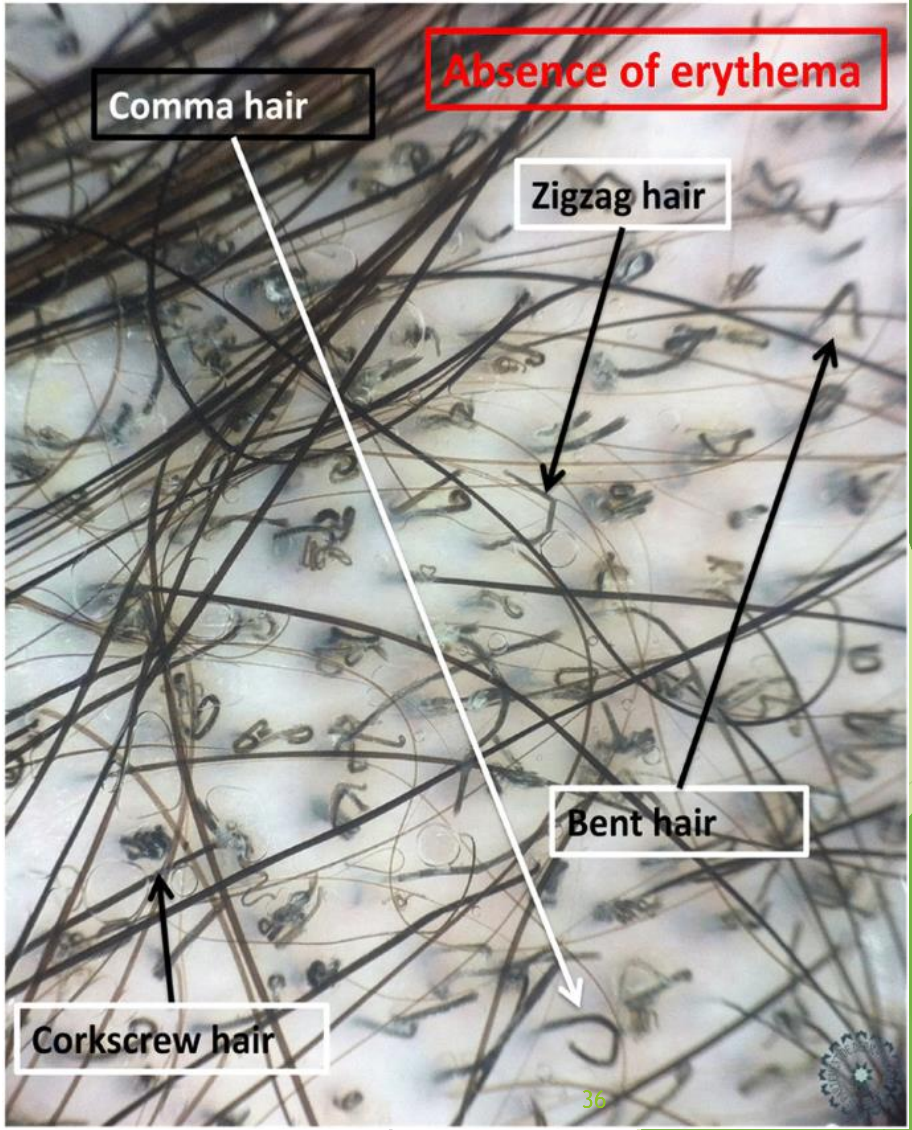
Kerion

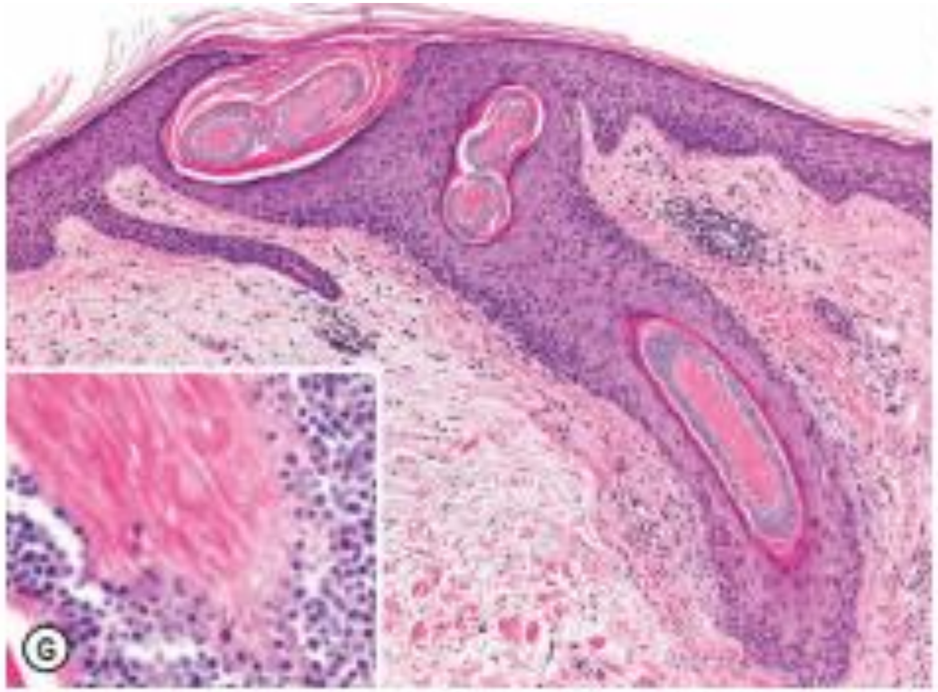
Fig. 77.14 Favus due to ***Trichophyton schoenleinii*****.** Scarring alopecia with erosions and several scutula on the occipital scalp. The latter represent masses of keratin plus fungi.
Courtesy, Israel Dvoretzky, MD.

Investigations

- Wood's light:- cats and dogs fight & growl sometimes
- **koh microscopy**:- hyaline hyphae [translucent, branched, septated & show pattern].
- Culture:- sabourauds dextroseagar 25-30 in 2-4 weeks.
- H.P:- staing → pas , fontana masson, GMC – Mayers's
- Dermoscopy :- comma, corkscrew, morsecode & zigzag.
- Favus → Perifollicular wax color

FIGURE 1: Videodermoscopy 20x, comma and corkscrew hair




G


F

DD:-

- AA, Trichotilomania,
- Scaring A → LPD, FD, DL & CCCA.
- Scalyscalp → SD & PS.
- Inflammatory→impetigo & bacterial folliculitis.

ttt

- Systemic R/ ultra griseoflavin.
- Topical therapy :- clinical cure / mycological cure

Tinea capitis in adult

- Rare → sebum, malassezia & good CMI
- Predisposing factor:- immunosuppressants, DM, anemia, steroid & hernia.
- Mode of transmission:- Direct autoinulation.
- Atypical presentation:- like : Eczema, SD, PS, AF & FD
- Prolonged ttt:- fungicidal drug →large dose & long duration.

Other tenias

- Faciei
- Barbae
- Corporis
- Manum
- Cruris
- Pedis
- Ungum

Faciei

- Glabrous skin.
- Children & females.
- T.rubram & T.mentagrophyte.
- Skin only & no follicles→Topical only
- Kitten & puppies → M.cannis
- Patch →upper lip & chin → itch & burn exacerbate after sun.
- Erythematous scaly patch any active border.
- DD:- S.D.Ps, rosetia, DLE, PLE & bownoid solar keratosis.

Fig. 77.11 Tinea faciei. A Erythema and scale on the nose and philtrum of a young child. The location and lack of central clearing may lead to the misdiagnosis of dermatitis with secondary impetiginization. **B** Child with pink papules, a few tiny pustules, and thin annular and arcuate scaly plaques in a perinasal and perioral distribution. These clinical findings could be mistaken for periorificial dermatitis. **C** Hyperpigmented lesions with subtle arcuate and annular configurations in a woman with darkly pigmented skin. There is minimal scale following the use of topical corticosteroids ("tinea incognito"). The clue to the application of topical

barbae

- Affect skin & follicle → beard & moustache.
- Adult male.
- **Zoophilic** (farm animals) →T.mentagraphyte & T.verrucosum.
- Papule, pustule & pustular folliculitis →Systemic ttt.
- Loose hair → easily removed without pain & lymphadenopathy.
- Scarring alopecia
- **Anthropophilic** → M.canis → less severe, superficial & reversible AA.

→Topical ttt.

Fig. 77.10 Superficial form of tinea barbae due to *Trichophyton rubrum*. Several follicular pustules are evident.
Courtesy, Jean L Bolognia, MD.

Corporis

- Trunk & extremities
- except hair, nail, palm & sole.
- T.rubraum & T.mentography →tropical regions.
- I.P. →1-3 wks.
- Well defined erethematous annular scaly with active egde.

- Variants:-

Tinea profunda:- deep in skin, dermis & SC tissue.

Majocchi granuloma:- deep folliculitis → T.rubrum.

Imbricata:- T.concentricum/anthropophilic/concentric rings/erythema gyralum rapens.

DD:- annular lesions.

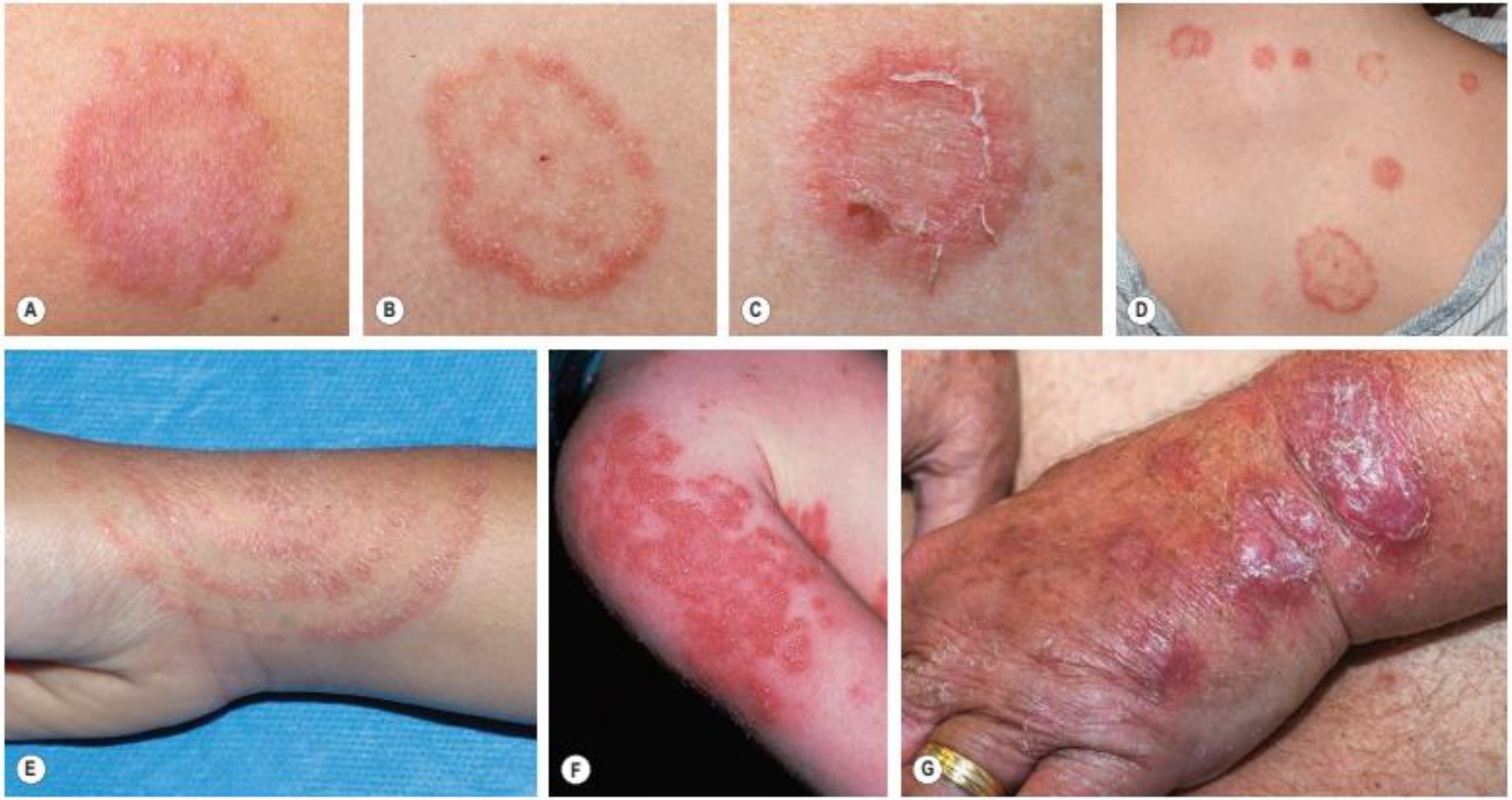


Fig. 77.6 Tinea corporis. A There is a subtle annular configuration with a border composed of individual, slightly scaly papules. **B** Classic annular lesion with a scaly raised border and central clearing. **C** Annular lesion with trailing scale reminiscent of erythema annulare centrifugum. **D** Multiple annular and circinate lesions of various sizes on the upper back. **E** Scaly concentric rings on the arm. **F** Pustules within multiple figurate lesions on the upper arm. **G** Inflammatory nodules on the dorsal hand and a thick granulomatous plaque on the distal forearm (tinea profunda). There is associated scale, including on the digits, and a few of the papulonodules are follicular (Majocchi granuloma). *B, D, Courtesy, Julie V Schaffer, MD. C, G Courtesy, Kalman Watsky, MD.*

Fig. 77.7 Nodular perifolliculitis (Majocchi granuloma). Perifollicular inflammation and follicular pustules on the leg due to *Trichophyton rubrum*. A few of the lowermost papules have a granulomatous appearance. *Courtesy, Kalman Watsky, MD.*

Manum

- -T.rubrum,T.mentographyte,merdigen & E.floccosum.
- Palm
- Diffuse hyperkeratosis.
- Autoinoculation.
- 2 feet 1 hand syndrome
- Exofoliating scaly vesicular patch.
- Under, ring & watch → poor peripheral circulation. PPK.
- Systemic ttt :as thick skin

Fig. 77.9 Tinea manuum. A Diffuse scaling of the palm of one hand, with accentuation in the creases. **B** Multiple collarettes of scale reminiscent of keratolysis exfoliativa on one palm.

Cruris

- Inguinal , behavioural hygiene.
- Obesity, DM & hyperhidrosis.
- Spared scrotum.
- Causes of intertrigoo infection → PS, MF, HH & Darier
- Early topical ttt is recommended.

Fig. 77.8 Tinea cruris. A thin, "broken-up" erythematous plaque with an arciform papular border on the upper inner thigh.

TINEA CRURIS: COMMON CAUSATIVE PATHOGENS	
Dermatophyte	**Clinical features**
Trichophyton rubrum	• Most common cause of tinea cruris • Infection tends to be chronic • Fungus not viable in scale (e.g. on furniture, rugs, linens) for long periods of time • Frequent extension to buttocks, waist, and thighs
Epidermophyton floccosum	• Commonly associated with "epidemics" of tinea cruris in locker rooms or dormitories • Infection is acute (rarely chronic) • Arthroconidia are viable in scale (e.g. on furniture, rugs, linens) for long periods of time • Infection rarely extends beyond groin region • Causative agent of "eczema marginatum" (well-demarcated borders with multiple small vesicles or, sometimes, vesiculopustules)
T. mentagrophytes (previously *T. mentagrophytes* var. *mentagrophytes*)	• Infection tends to be more severe and acute, with intense inflammation and pustule formation • May rapidly spread to the trunk and lower extremities, causing a severe inflammatory condition • Often acquired from animal dander

Pedis

- Mixed infection → dermatophyte, candida & bacteria.
- Mixed ttt → web space infection toe.
- PF:- moist environment & occlusive shoes.
- 4-5th toes.

❑ Clinical types

- Interdigital → most common, erythema, scaling, maceration, fissure, peeling.
- Ulcerative
- Vesiculobullous → T.mentophyte & medial foot.
- Moccasin type:- T.flocussure & T.rubrum → erythema, hyperkeratosis & silvery white scales.

THE FOUR MAJOR TYPES OF "TINEA PEDIS" CAUSED BY DERMATOPHYTES AND NON-DERMATOPHYTES			
Type	Causative organism	Clinical features	Treatment considerations
Moccasin	*Trichophyton rubrum* *Epidermophyton floccosum* *Neoscytalidium hyalinum* *N. dimidiatum*	Diffuse hyperkeratosis, erythema scaling and fissures on one or both plantar surfaces; frequently chronic and difficult to cure*; occasionally associated with immune deficiency	Topical antifungal plus product with urea or lactic acid; may require oral antifungal therapy
Interdigital	*T. interdigitale* (previously *mentagrophytes* var. *interdigitale*) *T. rubrum* *E. floccosum* *N. hyalinum* *N. dimidiatum* *Candida* spp. *Fusarium* spp.	Most common type; erythema, scaling, fissures and maceration in the web spaces; the two lateral web spaces are most commonly affected; "dermatophytosis complex" (fungal infection followed by bacterial invasion†) can develop; pruritus common; may extend to dorsum and sole of foot	Topical antifungal; may require topical or oral antibiotic if superimposed bacterial infection
Inflammatory (vesicular)	*T. mentagrophytes* (previously *T. mentagrophytes* var. *mentagrophytes*)	Vesicles and bullae on the medial foot; often associated with a dermatophytid reaction‡	Topical antifungal usually sufficient
Ulcerative	*T. rubrum* *T. interdigitale* (previously *T. mentagrophytes* var. *interdigitale*) *E. floccosum*	Typically an exacerbation of interdigital tinea pedis; ulcers and erosions in the web spaces; commonly secondarily infected with bacteria; seen in immunocompromised and diabetic patients	Topical antifungal; may require topical or oral antibiotics if secondary bacterial infection (common)

Dermatophytes — Non-dermatophytes

Fig. 77.15 Tinea pedis. A Diffuse scaling on both feet (moccasin type) as well as on the right hand, representing "one hand–two feet" tinea.
B Maceration between the third and fourth toes in the interdigital form. **C** Erythema, scale-crust, and bullae in the inflammatory form. **D** Extension of tinea pedis onto the dorsal foot in a serpiginous configuration of erythema and papules. Scale is minimal due to the patient's use of a potent topical corticosteroid for presumed dermatitis.
B, Courtesy, Jean L Bolognia, MD; D, Courtesy, Julie V Schaffer, MD.

Fig. 77.16 Extensive tinea corporis emanating from tinea manuum and tinea pedis. Note the confluent involvement, serpiginous scaly borders, and associated tinea unguium of the toenails and one fingernail.

Complications of T.pedis

- Bacterial superinfection.
- Ide reaction.
- Cellulitis.

- DD

1. Erythlasma.
2. Candida.
3. Bacterial.
4. PS.
5. Eczema
6. Syphilis
7. Juvenile plantar dermatis

Ttt:- topical usually sufficient → topical & oral Ab & oral antifungal.

ungum

- Dermatophytic onchomycosis.
- T-rubrum ,t- mentagrophyte , E- floccosum

Systemic ttt.

Fig. 77.17 Tinea unguium. Onycholysis, yellowing, crumbling and thickening of the fingernails (A), thumb nails (B), and toenails (C) in the distal/lateral subungual variant. D White discoloration of the toenail in the superficial white variant. E Hyphae within a formalin-fixed, PAS-stained nail plate.

D.D. of tinea cruris [intertrigo]

1. Candida.
2. Erythrasma.
3. Bacterial intertrigo.
4. Flexural PS.
5. M.F.
6. Contact dermatitis.
7. Heily heily
8. Darier

DD of tinea manum

1. PS, eczema.
2. Syphilis.
3. PRP.
4. PPK.

DIFFERENTIAL DIAGNOSES OF DERMATOPHYTE INFECTIONS				
Tinea corporis	**Tinea cruris**	**Tinea faciei**	**Tinea capitis**	**Tinea pedis**
Dermatitis • Nummular eczema • Atopic • Stasis • Contact • Seborrheic (petaloid) Tinea versicolor Pityriasis rosea Parapsoriasis Erythema annulare centrifugum Annular psoriasis Subacute lupus erythematosus Granuloma annulare Impetigo	Cutaneous candidiasis Intertrigo • Seborrheic dermatitis • Psoriasis Erythrasma Contact dermatitis Lichen simplex chronicus Parapsoriasis/mycosis fungoides Hailey–Hailey disease Langerhans cell histiocytosis	Dermatitis • Seborrheic • Perioral • Contact Rosacea Lupus erythematosus Acne vulgaris Annular psoriasis (children)	Seborrheic dermatitis Alopecia areata Trichotillomania Psoriasis If pustules: • Pyoderma • Folliculitis If scarring: • Lichen planus • Discoid lupus erythematosus • Folliculitis decalvans • Central centrifugal cicatricial alopecia	Dermatitis • Dyshidrotic • Contact Psoriasis • Vulgaris • Pustular Juvenile plantar dermatosis Secondary syphilis If interdigital: • Erythrasma • Bacterial infection, e.g. GNR

Dermatophytide
Ide reaction

- Def:- non-infective cutaneous eruption.
- Allergic resistance to distant dermatophytide reaction.

❑ Diagnostic orientation

1. Proven dermatophytide infection.
2. Distant eruption.
3. Spontaneous disappearance after ttt of dermatophytide.
4. Same morphology of aid types.

❑ Clinical types:-

1. Pompholyx like → palmar surface of finger.
2. Small follicular papules → trunk & limb.
3. Others → EN-EM, EA urticarial.

SUGGESTED SYSTEMIC REGIMENS FOR DERMATOPHYTOSES				
	Fluconazole	**Griseofulvin**	**Itraconazole***	**Terbinafine**
Tinea pedis (moccasin type)/ tinea manuum (adults)	150–450 mg/week × 4–6 weeks[27]	750–1000 mg/day (microsize) or 500–750 mg/ day (ultramicrosize) × 4 weeks	200–400 mg/day × 1 week	250 mg/day × 2 weeks
Tinea pedis (moccasin type)/ tinea manuum (children)	6 mg/kg/week × 4–6 weeks	15–20 mg/kg/day (microsize suspension) × 4 weeks	3–5 mg/kg/day (maximum 400 mg) × 1 week	Daily dosing as for tinea capitis (see below) × 2 weeks
Tinea unguium (adults)	Toenail ± fingernail involvement:			
	150–450 mg/week until nails are clear[27]	1–2 g/day (microsize) or 750 mg/day (ultramicrosize) until nails are normal†	200 mg/day × 12 weeks or 200 mg BID × 1 week/month for 3–4 consecutive months	250 mg/day × 12 weeks
	Fingernail involvement only:			
	150–450 mg/week until nails are clear[27]	1–2 g/day (microsize) or 750 mg/day (ultramicrosize) until nails are normal†	200 mg/day × 6 weeks or 200 mg BID × 1 week/month for 2 consecutive months	250 mg/day × 6 weeks
Tinea unguium (children)	6 mg/kg/week × 3–4 months (fingernails) or 5–7 months (toenails), or until nails are clear	20 mg/kg/day (microsize suspension) until nails are normal†	5 mg/kg/day (<20 kg), 100 mg/day (20–40 kg), 200 mg/day (40–50 kg), or 200 mg BID (>50 kg) × 1 week/month for 2 (fingernails) or 3 (toenails) consecutive months	62.5 mg/day (<20 kg), 125 mg/ day (20–40 kg) or 250 mg/day (>40 kg) × 6 weeks (fingernails) or 12 weeks (toenails)
Tinea corporis (extensive, adults)	150–200 mg/week × 2–4 weeks	500–1000 mg/day (microsize) or 375–500 mg/ day (ultramicrosize) × 2–4 weeks	200 mg/day × 1 week	250 mg/day × 1 week
Tinea corporis (extensive, children)	6 mg/kg/week × 2–4 weeks	15–20 mg/kg/day (microsize suspension) × 2–4 weeks	3–5 mg/kg/day (maximum 200 mg) × 1 week	Daily dosing as for tinea capitis (see below) × 1 week
Tinea capitis (adults)‡	6 mg/kg/day × 3–6 weeks	10–15 mg/kg/day (ultramicrosize; usually maximum 750 mg/day) × 6–8 weeks	5 mg/kg/day (maximum 400 mg) × 4–8 weeks	250 mg/day × 3–4 weeks§
Tinea capitis (children)‡	6 mg/kg/day × 3–6 weeks	20–25 mg/kg/day (microsize suspension) × 6–8 weeks	5 mg/kg/day (maximum 500 mg) × 4–8 weeks	Granules: 125 mg (<25 kg), 187.5 mg (25–35 kg) or

Other moulds

Tinea nigra

Piedra

Black piedra

White piedra

Tinea nigra

- Superficial fungal infection of the palm.
- Young adults
- Dematoceus fungi (hortaea werneckii)
- Solitary pigmented patch [well demarcated on palm].
- Asymptomatic & non scaly.
- Dark pigmented border.
- Palm mainly – sole & rare trunk and neck

- Microscopy
- Koh microscopy :- brown dermatiaceus branched septated hyphae.
- Culture:- slowly growing colony green black [yeast-like] or black [mould like]
- DD:-
- junctional melanolytic nevus.
- Hyperpigmented P.V.
- Addison
- Syphilis
- Silver nitrate stain

Ttt→topical azol atylan.


THANK YOU